AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM298
Operating Systems

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 19/Nov./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 1.5 Hours

Instructions:

- Total Marks: **30**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are NOT allowed.

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Part 1: Multiple Choice Questions: [5 marks]

Question 1: Choose the most correct answer.

1. What is an operating system?
   a. A collection of programs to manage hardware
   b. An interface between the hardware and application
   c. A system service provider to the application programs
   d. All the choices.
2. Whenever a process needs I/O to or from a disk it issues:
   a. A system call to the CPU.
   b. A special procedure
   c. A system call to the OS
   d. Any of the choices.
3. The _____ of a process contains the local variables.
   a. Text section.
   b. Data section.
   c. Stack.
   d. Heap.
4. _____: to keep track of which users use how much and what kind of computer resources.
   a. File Management
   b. Logging
   c. Security
   d. Memory Management
5. _____: provide(s) an interface to the services provided by an operating system:
   a. Device Driver
   b. Device Controller
   c. System call
   d. Both a and c

Part 2: Short Essay Questions: [15 marks]

Question 2: [2 marks]
What is Vectored Interrupt? Define.

Question 3: [2 marks]
Explain how **Direct Memory Access (DMA)** handles interrupts during data transfer, and how this differs from interrupt handling in low-speed devices.

Question 4: [3 marks]
What do the following acronyms stand for?
a- PAN
b- IPC
c- DLM

Question 5: [4 marks]
Error detection is one of the operating system services provided to support users. Explain its role and how it can improve the user's ability to effectively use the system.

Question 6: [4 marks]
Define **Linker** and **Loader** in Computer Science context.

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Part 3: Problem Solving: [10 marks]

Question 7: [4 marks]

What do the following Shell commands do?

a. gedit
b. ls -l
c. ps
d. chown

Question 8: [6 marks]

Write a **Bash script** that:

1. Prompts the user to **enter the number of grades to be processed**.
2. Accepts the grades one by one using a **loop**.
3. Counts and prints:
   - The number of **passing grades** (greater than or equal to 50)
   - The number of **failing grades** (less than 50)
4. Assume that all grades entered are **within the range 0–100**.

Remark: *You do not need to handle non-numeric input or out-of-range values.*

Expected Output:

```
Enter the number of grades to process: 5
Enter grade 1: 60
Enter grade 2: 45
Enter grade 3: 80
Enter grade 4: 35
Enter grade 5: 50
Number of passing grades: 3
Number of failing grades: 2
```

End of questions

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